

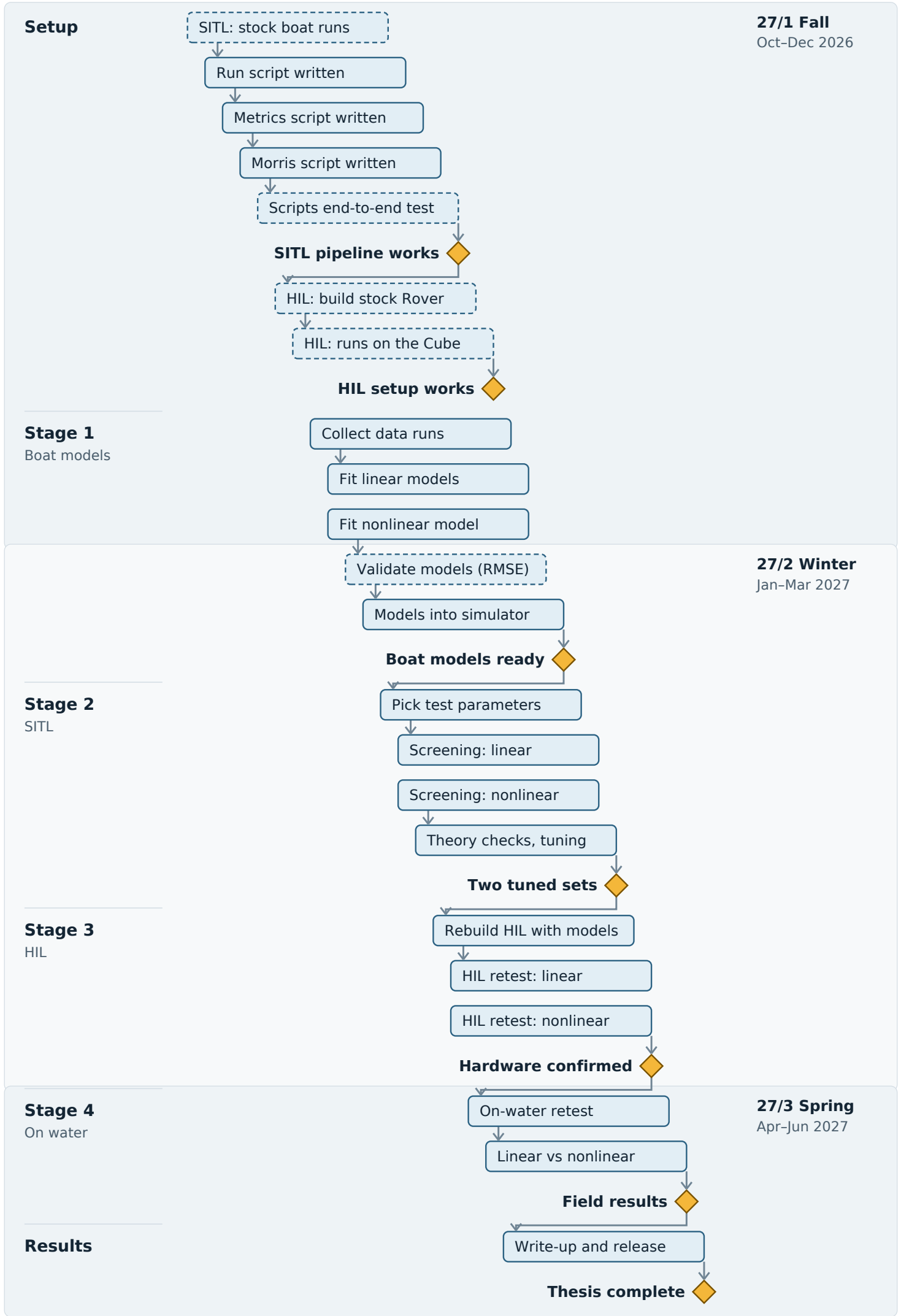
Tuning ArduPilot Speed and Heading Control for Small USVs

Research plan by NPS quarter, finishing June 2027. Platform: Blackjack 42, ArduPilot Rover, Cube Orange+.

Click any block or milestone for details. Each detail page links back here.

Milestones are gates; dashed blocks are validation checks; stacked blocks with the same start run in parallel.

Stage 1 data collection runs alongside setup.



[Validation check](#)[Setup](#)

SITL runs with ArduPilot's stock boat

Target: Oct 2026

What happens

Set up the build environment (Ubuntu or WSL), clone ArduPilot, and launch Rover SITL with the built-in motorboat frame. Connect Mission Planner to it.

Why

Confirms the simulator works before any custom boat model exists, so software problems are solved early.

Task

Setup

Run script

Target: Oct 2026

What happens

A Python script (pymavlink) that runs one test case: it loads a baseline parameter file, sets the parameters for that case, runs the maneuver (a speed step, a turn step, or a waypoint mission), and saves the log. It works the same way against SITL and against the Cube.

Why

Every test in Stages 2 and 3 goes through this script, so all runs are identical and need no manual steps.

[Task](#) Setup

Metrics script

Target: Oct-Nov 2026

What happens

Reads each log and computes the response measurements (rise time, settling time, overshoot, steady-state error, cross-track error, speed lost in turns) and the control-effort measurements (peak command, command rate, RMS effort).

Why

Every comparison in the thesis is based on these numbers, including response versus control effort.

Task

Setup

Morris screening script

Target: Nov 2026

What happens

Uses the Morris method (through the SALib library) to generate the list of parameter combinations to test, sends each one to the run script, collects the measurements from the metrics script, and ranks the parameters by their effect on response and on control effort. It also plots response against control effort to find the best balance.

Why

This is the script that performs the Stage 2 screening: it determines which speed and steering parameters matter most.

[Validation check](#)[Setup](#)

Scripts end-to-end test

Target: Nov 2026

What happens

Run a small screening batch (about ten cases) on the stock boat through all three scripts, with no manual steps, including the same baseline case three times. Check that parameters are set and reset correctly, every log and measurement is correct, repeated runs agree (their spread sets the noise floor), sped-up runs match real-time runs, and a failed run does not stop the batch.

Why

Proves the whole chain works before it is trusted with the full overnight screenings.

Milestone Setup

Milestone: SITL pipeline works

Target: Nov 2026

What happens

Automated testing works end to end on the stock boat.

Why

From here, the boat models can be swapped in without changing anything else in the pipeline.

[Validation check](#)[Setup](#)

HIL: build the stock Rover firmware

Target: Nov 2026

What happens

Build Simulation on Hardware firmware with ArduPilot's sitl-on-hw.py script: vehicle Rover, simulator class SimRover, board Cube Orange+, using the unmodified rover model.

Why

Ready-made Simulation on Hardware firmware exists only for Copter. This confirms the Rover version compiles and fits in the Cube's memory.

[Validation check](#)[Setup](#)

HIL: firmware runs on the Cube

Target: Nov-Dec 2026

What happens

Flash the firmware and reload the parameter file (flashing resets parameters). Then run the same run script used for SITL, connected to the Cube over USB instead of to SITL. Setting SIM_OH_MASK lets the real servo and ESC move; the propeller is removed.

Why

Confirms the scripts work unchanged on the real autopilot, so Stage 3 can reuse them.

Milestone Setup

Milestone: HIL setup works

Target: Dec 2026

What happens

Simulation on Hardware runs for Rover on the Cube Orange+ and accepts the test scripts.

Why

Removes the largest technical risk before the research stages begin.

Task

Stage 1 · Boat models

Collect data runs on the water

Target: Nov-Dec 2026, alongside setup

What happens

In Manual mode: throttle steps across the full speed range (from rest and between speeds), coast-downs from several speeds including top speed, rudder steps of $+\delta_0$ and $-\delta_0$ at two or three speeds, and the safe turning envelope (progressively sharper turns while logging roll). At least three repeats per condition. Wind and current are recorded, and an RC throttle cut is kept ready.

Why

This data builds both boat models and sets the safe turning limit used in every later test. It does not depend on the software setup, so it runs alongside it to save time. The simulation has no roll dynamics, so capsize limits must be measured on the water.

Task

Stage 1 · Boat models

Fit the linear models

Target: Dec 2026

What happens

First-order speed and Nomoto turning models, fitted from the step tests: a graphical estimate first (K^* from the steady-state value, τ from the 63.2% crossing), then a least-squares fit with `lsqcurvefit`. Speed gains and time constants are found at each throttle level, turning gains and time constants at each speed.

$$\frac{du}{dt}(t) = \frac{K_u^* \delta_T(t) - u(t)}{\tau_u}$$

$$\frac{dr}{dt}(t) = \frac{K_r^* \delta_R(t) - r(t)}{\tau_r}, \quad \frac{d\psi}{dt}(t) = r(t)$$

$$y(t) = K^* \delta_0 (1 - e^{-t/\tau}), \quad K^* = \frac{y_\infty}{\delta_0}, \quad y(\tau) = 0.632 y_\infty$$

Why

The linear models provide the control-theory predictions and act as the linear boat in simulation.

Task

Stage 1 · Boat models

Fit the nonlinear speed model

Target: Dec 2026

What happens

Speed model with thrust, linear-plus-quadratic drag, and a turn-drag term. The drag coefficients come from the coast-downs and the thrust coefficient from the steady speeds. A dead zone or a drag lookup table is added if the data show one. Runs in parallel with the linear fit.

$$\frac{du}{dt}(t) = c_T \delta_T(t) - c_1 u(t) - c_2 |u(t)| u(t) - c_r |r(t)| u(t)$$

Why

One model that is accurate across all speeds, used as the nonlinear boat in simulation.

Validation check

Stage 1 · Boat models

Validate the models (RMSE)

Target: Jan 2027

What happens

Replay runs not used in the fit through each model and overlay the prediction on the measured response. Report the RMSE for each model, compare linear and nonlinear across speeds, and check whether the linear gain changes with throttle level.

Why

Every simulation result depends on the models, so they must be shown to predict runs they have never seen.

Task

Stage 1 · Boat models

Put the models into the simulator

Target: Jan 2027

What happens

Edit ArduPilot's rover simulator (`libraries/SITL/SIM_Rover.cpp`) to contain both boat models, with a simulator setting that chooses which one the simulated boat uses, so SITL and HIL can switch between them without rebuilding. Both use speed-dependent Nomoto turning, plus the measured sensor noise and actuator delay.

Why

The same edited file runs in SITL and on the Cube, so both stages use the same boat.

Milestone

Stage 1 · Boat models

Milestone: boat models ready

Target: Jan 2027

What happens

Validated linear and nonlinear boat models running inside ArduPilot's simulator.

Why

Stage 2 screening can start.

Task

Stage 2 · SITL

Pick the test parameters

Target: Jan 2027

What happens

Select the parameters to screen: the speed and steering controller parameters, plus the motor-output settings that shape the commands sent to the motor and rudder (such as the throttle slew rate and steering scaling with speed) and the navigation settings that shape speed and turns during waypoint missions (such as the waypoint radius and turn radius). Each gets a realistic low-to-high tuning range; the turning-limit parameter is capped at the measured safe envelope.

Why

The motor-output and navigation settings also change how the boat holds speed and turns, especially in missions and in the cornering study. The screening only varies parameters within their ranges, so the list and ranges come first.

Task Stage 2 · SITL

Screening with the linear model

Target: Feb 2027

What happens

Morris elementary-effects screening: random starting points, each trajectory changing one parameter at a time, $r(k + 1)$ automated runs in total. For every measurement, each parameter gets an importance score (μ^*) and an interaction score (σ).

$$\mu_i = \frac{1}{r} \sum_{j=1}^r EE_i^{(j)}, \quad \mu_i^* = \frac{1}{r} \sum_{j=1}^r |EE_i^{(j)}|, \quad \sigma_i = \sqrt{\frac{1}{r-1} \sum_{j=1}^r (EE_i^{(j)} - \mu_i)^2}$$

Why

Ranks many parameters at low cost and flags interactions that one-at-a-time testing would miss.

Task

Stage 2 · SITL

Screening with the nonlinear model

Target: Feb 2027

What happens

The same screening, run with the nonlinear boat model. Runs in parallel with the linear screening.

Why

Comparing the two rankings shows whether the choice of model changes which parameters matter.

Task

Stage 2 · SITL

Theory checks and tuning

Target: Feb-Mar 2027

What happens

Targeted runs test each control-theory prediction, for example the delay a filter adds or the slope a rate limiter imposes. Parameters are flagged as redundant, masked, or ineffective. The prioritized set and its values are chosen at the best balance between response and control effort, and the turning settings that carry the most speed within the safe envelope are found.

Why

Turns the rankings into the thesis results: a tuning order, a theory correspondence, and cornering guidance.

Milestone

Stage 2 · SITL

Milestone: two tuned sets

Target: Mar 2027

What happens

Rankings, theory results, and tuned parameter sets from both models.

Why

Stage 3 confirms these results on real hardware.

Task

Stage 3 · HIL

Rebuild HIL with the boat models

Target: Mar 2027

What happens

Rebuild the Simulation on Hardware firmware with the edited simulator file, flash it, and reload the parameter file.

Why

Puts the same boats used in SITL onto the real autopilot.

Task

Stage 3 · HIL

HIL retest with the linear model

Target: Mar 2027

What happens

On the real Cube with the real servo and ESC: the top-ranked parameters, plus a few low-ranked ones as a check, at low, default, and high values; the linear-tuned set on the same maneuvers and mission; and a spot check of the theory predictions most affected by hardware (filters, rate limits). Runs in real time.

Why

Confirms that the real autopilot hardware does not change the SITL results.

Task

Stage 3 · HIL

HIL retest with the nonlinear model

Target: Mar 2027

What happens

The same retest with the nonlinear model and the nonlinear-tuned set. Runs in parallel with the linear retest.

Why

Confirms the nonlinear results on real hardware.

Milestone

Stage 3 · HIL

Milestone: hardware confirmed

Target: Mar 2027

What happens

SITL findings hold on the real autopilot, or the differences are measured and explained.

Why

Stage 4 can test on the water with confidence in the settings.

Task

Stage 4 · On water

On-water retest

Target: Apr 2027

What happens

The top-ranked parameters, plus a few low-ranked ones, retested on the real boat with the same maneuvers and waypoint mission and a fixed number of repeats. Turning settings are raised in steps within the safe envelope. The control-theory predictions are checked on the water. Safety settings stay on throughout.

Why

Shows whether the simulation rankings hold on the real boat; rankings from SITL, HIL, and the water are compared with the Spearman rank correlation.

Task

Stage 4 · On water

Linear-tuned versus nonlinear-tuned boat

Target: Apr-May 2027

What happens

The linear-tuned and nonlinear-tuned parameter sets each run the same waypoint mission, with repeats and alternating order to balance wind and current. Compared on cross-track error, heading error, speed through turns, mission time, and control effort.

Why

Measures how much a nonlinear boat model improves the real boat's performance.

Milestone

Stage 4 · On water

Milestone: field results

Target: May 2027

What happens

On-water rankings, theory checks, and the linear-versus-nonlinear comparison.

Why

All data needed for the thesis is collected.

Task

Results

Write-up and open-source release

Target: May-Jun 2027 (chapters drafted along the way)

What happens

Chapters are drafted along the way: background in Fall, methods in Winter, results in Spring. The final write-up reports the prioritized parameters and tuning order, the parameter-to-control-element correspondence, the cornering guidance, the simulation-to-water comparison, and the linear-versus-nonlinear results. Release the identification scripts, simulator model, test scripts, and HIL setup on GitHub.

Why

Delivers the thesis and lets other users repeat the process on their own boats.

Milestone Results

Milestone: thesis complete

Target: Jun 2027

What happens

Thesis approved and tools released.